

DENTAL EMERGENCIES

Common Dental Emergencies that present to the CE include:

- A. Odontogenic Infections
- B. Dental Injuries
- C. Others

A. ODONTOGENIC INFECTION

1. Alveolar Abscess
 - Possible sources: Decayed teeth (most common), dental anomalies (i.e. fractured tubercles of dens evaginatus tooth most commonly premolars) as the fractures can result in pulp (nerve in the tooth) infection and abscess formation in teeth without any sign of decay
 - Symptoms: Fever, malaise, pain, facial cellulitis
 - Treatment:
 - I&D of fluctuant swellings, drainage of pus via root canal/socket, extraction
 - Antibiotics (amoxicillin + metronidazole or amoxicillin/clavulanic acid)
 - Supportive-antipyretics, analgesia, antiseptic mouth wash (Chlorhexidine)
2. Ludwig Angina
 - A dental infection from a lower posterior tooth could potentially result in this life-threatening cellulitis (of the submandibular, sublingual and submental spaces) resulting in airway compromise
 - Admission typically required for I&D, IV antibiotics and hydration
3. Primary Herpetic Gingivostomatitis (Herpes Simplex Type 1 infection)
 - Most common cause of severe oral ulceration in children
 - Most common age: Toddlers
 - Consider immune status (if immunocompromised, consider anti-viral course)
 - Treatment:
 - Symptomatic relief – analgesia, anti-pyretics, ice chips, ice cream, soft diet. If topical LA is prescribed, advise parents to monitor their child closely to prevent them from biting on their numbed lips/tongue as this can cause further pain and ulceration.
 - Chlorhexidine mouth wash (non-alcoholic preparation)
 - Review; self-limiting 10-14 days

B. DENTAL TRAUMA

1. Maxillofacial fractures (less common in children)
 - Assess for TMJ injuries (check range of motion, trismus, pain when opening/closing mouth, derangement of occlusion).
 - Refer OMS for further investigations – may require plating of bony fractures
2. Dental injuries
 - Most commonly affected teeth: Upper central incisors
 - Most common age:
 - Primary Dentition: Toddlers (when they are learning to walk)

References: International Association of Dental Traumatology (IADT) Guidelines for the Management of Traumatic Dental Injuries (2020): 1. Fractures and Luxations 2. Avulsion of Permanent Teeth 3. Injuries in the Primary Dentition

Disclaimer: This document reflects what is currently regarded as safe practice. However, as in any clinical situation, there may be factors which cannot be covered by a single set of guidelines. This document does not replace the need for the application of clinical judgement to each individual presentation.

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- Permanent Dentition: >6-7 years old (Often in children/teenagers involved in contact sports)
- For all injuries: Advise soft diet, good oral hygiene, chlorhexidine mouth rinse (non-alcoholic preparation), analgesics, antibiotics if deemed appropriate
- Assess for concomitant soft tissue injuries which may require suturing under LA and sedation if indicated
- Injuries should be referred to the dental clinic for further review to monitor for possible consequences of injury to the teeth such as tooth discoloration, pain/swelling and damage to permanent successor (applicable to primary teeth injuries) as well as continued growth and development of the affected tooth/teeth.
- Please refer to Appendix 1 for the management strategy for the various injuries

C. OTHER CONDITIONS REQUIRING REFERRALS TO ON CALL-DO/PAEDIATRIC DENTIST/OMS INCLUDE...

1. Torn frenulum/intra-oral injury (palate) in a very young child – also consider if this is true injury or abuse?
2. Repositioning and plating of jaw fractures
3. Suturing of lacerations, including the tongue
4. Splinting of/replacing knocked out teeth
5. Repositioning of displaced teeth
6. Repair broken teeth
7. Maxillofacial trauma – RTA, sports related (can subsequently fabricate mouth guards to protect from future injuries)
8. Any suspicion of oral or jaw tumours/cysts

During office hours, refer to NUCOHS (walk-in) with a referral letter – but patient may need readmission to CE if sedation is required.

For out of office hour referrals, contact On-call Dental officer or Registrar on duty.

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APPENDIX 1. DENTAL TRAUMA MANAGEMENT		
TYPE OF INJURIES	PRIMARY TEETH MANAGEMENT	PERMANENT TEETH MANAGEMENT
Concussion & Subluxation: Tooth in socket and not displaced, may have bleeding around gums and slight increased mobility	- Soft diet, analgesia - For permanent teeth with excessive mobility (2-3mm each direction), then reposition and stabilization is recommended	
Luxation: Tooth is displaced out of the socket (either towards the lip or towards the back of the throat)	- If bite unaffected → leave alone for spontaneous repositioning - If bite affected or tooth is very mobile posing a risk of aspiration → extract (may require sedation at CE)	- Reposition tooth and splint for stabilization - Suture gingival laceration if needed
Extrusion: Partial displacement of the tooth out of its socket	- If mild extrusion/bite unaffected → leave alone - If extrusion >3mm/excessively mobile → extract (may require sedation at CE)	
Intrusion: Tooth pushed into the gums	- Majority of cases to leave alone to re-erupt if tooth is present. - In severe intrusion (palpable) at the nasal area) and/or with crushing injuries, tooth may need to be extracted (rare)	- Reposition the tooth into the correct position (if intrusion ≥3mm) and splint for stabilization - Spontaneous eruption for mild intrusion (<3mm) in a young permanent tooth (age dependent usually 6-7 years old)
Avulsion: “Knocked out” tooth	DO NOT REIMPLANT primary/baby teeth To find out if tooth is permanent or primary - Permanent teeth are larger and yellower compared to the neighbouring milk teeth - Usually erupt at 6-7 years - Ask parents if front teeth have “fallen out” before	3. Over the phone advice for parents if tooth is knocked out: 1. Locate the tooth (if tooth is missing, consider aspiration/ embedded in soft tissue- radiographs/chest X-rays may be helpful to locate tooth) 2. Hold the tooth by the crown only (root still has viable cells on surface) – preservation of tooth vitality for continued root development

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		3. If debris present, gently rinse with saline (DO NOT SCRUB) 4. Place the tooth back into the empty socket (best option) or in the child's mouth (not advised if child unable to hold it without accidentally swallowing it) or in fresh milk. Other storage agents include saline or Hanks Balance Salt Solution 5. Visit A&E immediately 4. Arrival at <u>CE</u> ○ Immediate referral to dental for re-implantation and splinting (Prognosis declined after 60 minutes) ○ Tetanus if deemed necessary
Root fracture	• If stable → leave alone and monitor • If excessively mobile with risk of aspiration → extract coronal segment. Leave apical (root) segment if unable to retrieve. The residual root will resorb on its own.	• If tooth mobile, reposition tooth and splint for stabilization
Crown fracture: Crown (visible) part of the tooth is fractured	• To account for missing tooth fragment (to rule out that the fragment did not get embedded in the soft tissue) • Do NOT throw tooth fragment away as it may be reattached later by the dentist. • If no pulp exposure → refer to dentist to put bandage on tooth as the fracture site can be a portal of entry for infection. • If pulp exposure noted → refer to dentist for pulp treatment or extraction	

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